MILITARY SPECIFICATION SHEET

SPARK PLUG, SHIELDED, AIRCRAFT RECIPROCATING ENGINE
DETAILED SPECIFICATION FOR

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the spark plugs described herein shall consist of this document and the latest issue of specification MIL-S-7886.

1. SCOPE

1.1 Scope, This specification covers spark plugs for specific aircraft-engine combinations (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

oC 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-7886	Spark Plug, Shielded. Aircraft Reciprocating Engine, General Specification For.

3. REQUIREMENTS

3.1 Qualification. The spark plugs furnished under this specification shall conform to MIL-S-7886 and shall be a product which has been tested, and passed the qualification tests specified in MIL-S-7886 and the engine-airframe flight application tests specified herein, and has been listed on, or approved for listing on the applicable qualified products list.

3,2 Spark gap. Fine wire spark plugs for this application shall be gapped at a nominal .016 inch (.015 inch go .018 inch no go). Massive electrode spark plugs for this application shall be gapped at a nominal .016 inch (.015 inch go .019 inch no go).

oC Denotes changes

FSC 2925

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4. QUALITY ASSURANCE PROVISIONS

4.1 Engine/airframe flight test.

a. Test plugs will be operated on typical unit missions.

b. A test lot will constitute a minimum of 20 engine sets of spark plugs.

c. Any unscheduled spark plug removal where an engine malfunction is corrected by spark plug replacement constitutes a failure. Failures will be classified in two categories :

(1) Material failure: Material failure constitutes a structural failure of the plug or any component of the plug.

(2) Operational failure: An operational failure is a plug malfunction for any reason other than a structural failure.

d. The total failure rate for the first 400 hours shall not exceed 20 percent. Material failure shall not exceed 2 percent during this period. This 2 percent shall include any structural failures detected during reconditioning after the first 400 hours of operation.

e. Massive type spark plug tests will be limited to 400 hours.

f. Fine wire spark plugs shall be reconditioned after 400 hours and subjected to additional testing. Any unscheduled removals for other than material failure shall be reconditioned. A minimum of 90 percent of the original test lot shall be serviceable after being reconditioned.

g. The reconditioned plugs will be installed for an additional 400 hour test. The total failure rate for this test shall not exceed 30 percent with the material failures limited to 5 percent of the remaining test lot.

h. Whenever new and/or radical designs are submitted for qualification, the responsible Air Force engineering office shall have the prerogative of requiring a 150 hour flight test on four engine sets to determine if further large scale testing is feasible.

i. At the beginning of the test and every 100 hours thereafter, the following three tests, if and when required by the responsible engineering office, will be successfully accomplished three times :

(1) Engines shall be allowed to idle in auto-rich for 45 minutes (using normal defouling procedures) and shall be capable of producing acceptable torque at maximum power, that is, wet take-off or dry take-off power as applicable.

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(2) Test aircraft shall be flown at maximum rate of climb from point of takeoff to maximum operating altitude with satisfactory plug performance throughout. Upon reaching maximum operating altitude, the aircraft shall be operated at cruise power for ten minutes to cool the engine and then at maximum power for five minutes with satisfactory spark plus operation throughout. The aircraft shall be operated at maximum endurance power for 30 minutes with satisfactory spark plug operation,

(3) Three cold starts shall be attempted on a minimum of two engines. using current normal cold starting procedures, during temperatures of -10° Fahrenheit or lower with acceptable starting results. Each engine and plugs shall be allowed to "cold soak" for a minimum of 20 hours prior to each attempted start.

5. Requirements for preservation, packaging. packing and marking are specified in MIL-S-7886.

6. NOTES

6.1 Intended use. The spark plugs are intended for use in all aircrait-eng ine combinations using the R1830, R2000, R2800, . R3350 and R4360 engines.

6.2 With respect to products requiring qualification, awards will be made only for products which are. at the time set for opening of bids, qualified for inclusion in Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements , and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible Lo be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is San Antonio ALC/MMEDO, Kelly AFB, TX 78241 and information pertaining to qualification of products may be obtained from that activity.

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